

Nervous System, Special Senses & Renal

Nervous System, Special Senses & Renal Physiology

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1

ORGANISATION OF THE NERVOUS SYSTEM

The nervous system is the body's primary network for communication, integration, and control. With over 100 billion neurons and even more supporting glial cells, it coordinates perception, movement, homeostasis, cognition, and behaviour. It has two structural components (central and peripheral) and several functional subdivisions.

Division	Components	Function
Central nervous system (CNS)	Brain and spinal cord	Integration and processing; decision making; higher functions
Peripheral nervous system (PNS)	12 cranial and 31 spinal nerve pairs, ganglia, plexuses	Carries sensory input to and motor output from the CNS
Somatic NS	Corticospinal tract, dorsal columns, alpha-motor neurons	Voluntary movement and conscious sensation
Autonomic NS	Sympathetic, parasympathetic, enteric	Involuntary control of viscera, glands, smooth and cardiac muscle
Enteric NS	Myenteric and submucosal plexuses	The brain in the gut; largely autonomous control of GI motility and secretion

Afferent versus efferent

Afferent (sensory) fibres carry signals from the periphery towards the CNS; their cell bodies sit in the dorsal root ganglion (spinal) or cranial nerve ganglia. Efferent (motor) fibres carry signals from the CNS to the periphery. Somatic efferents run directly to skeletal muscle; autonomic efferents use a two-neuron chain (preganglionic to ganglion to postganglionic to effector).

2

NEURONS, GLIA & THE ACTION POTENTIAL

2.1 Neuron structure

Part	Function
Cell body (soma)	Protein and neurotransmitter synthesis (Nissl substance); integrates inputs; trophic centre
Dendrites	Receive synaptic inputs; summate EPSPs and IPSPs as graded potentials
Axon hillock	Action potential initiation site (highest density of voltage-gated Na channels)
Axon	Conducts the action potential; supports axoplasmic transport in both directions
Axon terminal (bouton)	Releases neurotransmitter into the synaptic cleft on AP arrival

Neurons are classified by shape: multipolar (most CNS neurons and motor neurons), bipolar (special sense organs such as retina and cochlea), and pseudounipolar (all primary somatic sensory neurons in the dorsal root ganglia).

2.2 Nerve fibre classification (Erlanger-Gasser)

Fibre	Myelination	Velocity	Function
A-alpha	Heavy	70 to 120 m/s	Proprioception (Ia, Ib); alpha-motor to muscle
A-beta	Myelinated	30 to 70 m/s	Touch, vibration, pressure
A-gamma	Myelinated	15 to 30 m/s	Motor to muscle spindle (intrafusal)
A-delta	Light	5 to 30 m/s	Sharp first pain; cold; crude touch
B	Light	3 to 15 m/s	Autonomic preganglionic fibres
C	None	0.5 to 2 m/s	Burning second pain; warmth; postganglionic sympathetic

Local anaesthetic order of blockade

Local anaesthetics block sodium channels and affect smaller fibres first: C (pain, sympathetic), then A-delta (acute pain, cold), then A-beta (touch, pressure), then A-alpha (motor, proprioception). This is why spinal or epidural patients lose pain first and motor power last, with recovery in reverse order.

2.3 Glial cells

Glial type	Location	Key functions
Astrocytes	CNS	Blood-brain barrier endfeet; potassium buffering; glutamate uptake; scar formation
Oligodendrocytes	CNS	Myelinate multiple CNS axons; destroyed in multiple sclerosis
Schwann cells	PNS	Myelinate one internode of one PNS axon; guide nerve regeneration
Microglia	CNS	Resident macrophages; phagocytosis; synaptic pruning; neuroinflammation
Ependymal cells	Ventricles / central canal	Produce and circulate CSF; form the blood-CSF barrier

The blood-brain barrier

Tight junctions between cerebral capillary endothelial cells, reinforced by astrocyte endfeet and pericytes, restrict large, charged, and pathogenic substances from the CNS. Oxygen, carbon dioxide, water, and lipid-soluble drugs cross freely; glucose (GLUT1) and amino acids need specific transporters; most antibiotics, chemotherapy agents, and large proteins cannot cross unless the barrier is disrupted.

2.4 The action potential

The resting membrane potential is about -70 mV (inside negative), set by unequal ion distribution and selective permeability. At rest the membrane is most permeable to potassium through leak channels, so potassium (equilibrium -94 mV) is the main determinant. The Na/K-ATPase (3 Na out, 2 K in) maintains the gradients and adds a small electrogenic contribution.

Phase	Membrane potential	Ion movement / channel state
Resting	-70 mV	Potassium leak maintains RMP; gated channels closed
Depolarisation	-70 to +30 mV (threshold ~-55)	Voltage-gated Na channels open, Na rushes in
Repolarisation	+30 to -70 mV	Na channels inactivate; voltage-gated K channels open, K flows out
After-hyperpolarisation	-70 to -80 mV	K channels close slowly; end of the refractory period

Refractory periods and saltatory conduction

Absolute refractory period: no action potential is possible because Na channels are inactivated; this enforces one-way propagation and caps firing frequency. Relative refractory period: a stronger stimulus can fire because Na channels have recovered but the membrane is still hyperpolarised. In myelinated fibres the impulse jumps node to node (saltatory conduction), which is faster and more energy efficient.

3

SYNAPSES, BRAIN & SPINAL CORD

3.1 Chemical synaptic transmission

- The action potential reaches the terminal and depolarises the membrane.
- Voltage-gated calcium channels open; calcium influx is the trigger for exocytosis.
- Calcium binds synaptotagmin; SNARE proteins drive vesicle fusion and neurotransmitter release into the 20 to 50 nm cleft.
- Neurotransmitter binds postsynaptic receptors, opening ion channels (ionotropic) or activating cascades (metabotropic GPCRs).
- Transmitter is cleared by reuptake (SERT, DAT, NET, EAAT), enzymatic degradation (AChE, MAO/COMT), or diffusion.

3.2 Key neurotransmitters

Transmitter	Action	Notable pathways / notes
Glutamate	Major excitatory; NMDA critical for plasticity (LTP)	Corticospinal, thalamocortical, hippocampal
GABA	Major inhibitory (GABA-A raises chloride)	Benzodiazepine and barbiturate sites; cortical interneurons
Acetylcholine	Excitatory at NMJ; arousal, attention, memory	NMJ, autonomic ganglia, basal forebrain (memory)
Dopamine	Reward, motivation, movement	Mesolimbic, mesocortical, nigrostriatal (depleted in Parkinson)
Serotonin	Mood, sleep, appetite, GI motility	Raphe nuclei; 90% of body serotonin is in the gut
Noradrenaline	Alertness, attention, fight-or-flight	Locus coeruleus; sympathetic postganglionic

3.3 The brain: gross anatomy

Region	Key structures	Functions
Cerebrum	Cortex, white matter, basal ganglia, hippocampus, amygdala	Cognition, language, voluntary movement, sensation, memory, emotion
Diencephalon	Thalamus, hypothalamus, epithalamus (pineal)	Sensory relay; homeostasis and ANS control; circadian rhythm
Midbrain	Colliculi, substantia nigra, red nucleus, CN III/IV	Visual and auditory reflexes; dopamine; pain modulation
Pons	Locus coeruleus, raphe, CN V to VIII	Relay to cerebellum; respiratory rhythm; cranial nerve functions
Medulla	Cardiovascular, respiratory, vomiting centres, CN IX to XII	Vital autonomic centres; sensory and motor decussation
Cerebellum	Purkinje cells, deep nuclei, peduncles	Coordination, balance, motor timing and learning (ipsilateral)

3.4 Cerebral cortex: functional areas

Area	Function	Deficit if damaged
Primary motor (BA4)	Voluntary movement; homunculus; origin of CST	Contralateral hemiparesis; UMN signs
Primary somatosensory (BA1-3)	Conscious somatosensation	Contralateral loss of fine touch and proprioception
Primary visual (BA17)	Conscious vision	Contralateral homonymous hemianopia (macular sparing)
Broca (BA44-45)	Speech production	Non-fluent expressive aphasia, comprehension intact
Wernicke (BA22)	Speech comprehension	Fluent but meaningless speech, poor comprehension
Prefrontal cortex	Executive function, personality, working memory	Personality change, poor planning and impulse control

Cortical lateralisation

Language is left-hemisphere dominant in about 90% of right-handers; the arcuate fasciculus links Broca and Wernicke areas (its damage gives conduction aphasia). Visuospatial processing is right-hemisphere dominant, so right parietal lesions cause left spatial neglect. Motor and sensory representation is contralateral; cerebellar representation is ipsilateral.

3.5 The limbic system

Structure	Function	Lesion effect
Hippocampus	Declarative memory formation; spatial navigation	Anterograde amnesia (bilateral)
Amygdala	Fear processing, emotional salience, aggression	Kluver-Bucy syndrome (bilateral)
Mammillary bodies	Memory relay (Papez circuit)	Wernicke-Korsakoff (thiamine deficiency), confabulation
Hypothalamus	Homeostasis, feeding, temperature, neuroendocrine control	Temperature and hormonal dysregulation, obesity

3.6 The basal ganglia

The basal ganglia are subcortical nuclei that select and refine voluntary movement, facilitating desired movements and suppressing unwanted ones. The striatum (caudate plus putamen) is the main input; the globus pallidus interna is the main inhibitory output to the thalamus; the subthalamic nucleus and substantia nigra tune the circuit.

Direct versus indirect pathway

Direct pathway (facilitates movement): cortex to striatum (D1) to GPi (inhibited) to thalamus (disinhibited) to cortex. Dopamine excites D1 and promotes movement.

Indirect pathway (suppresses movement): cortex to striatum (D2) to GPe to STN to GPi to thalamus (inhibited). Dopamine inhibits D2 and so reduces suppression.

Parkinson disease: loss of nigral dopamine tips the balance towards inhibition of the thalamus, causing bradykinesia and rigidity (treated with L-DOPA). Huntington disease: loss of indirect-pathway striatal neurons causes chorea.

3.7 Spinal cord and tracts

The spinal cord has central H-shaped grey matter (cell bodies) surrounded by white matter (myelinated tracts). The dorsal horn handles sensory relay, the lateral horn (T1 to L2) holds sympathetic preganglionic neurons, and the ventral horn holds alpha and gamma motor neurons.

Tract	Modality / function	Decussation
Dorsal column-medial lemniscus	Fine touch, vibration, proprioception	In the medulla
Spinothalamic (anterolateral)	Pain, temperature, crude touch	In the cord (within 1 to 3 segments)
Spinocerebellar	Unconscious proprioception	Does not decussate (ipsilateral)
Lateral corticospinal	Voluntary fine motor (distal limbs)	Pyramidal decussation in the medulla (~85%)
Vestibulospinal	Antigravity tone and posture	Does not decussate

Brown-Sequard syndrome (cord hemisection)

Below the lesion, the same (ipsilateral) side loses motor power (corticospinal, still uncrossed) and fine touch and proprioception (dorsal columns, still uncrossed). The opposite (contralateral) side loses pain and temperature (spinothalamic, already crossed in the cord). At the level of the lesion there is a band of ipsilateral loss of all modalities.

4

AUTONOMIC NERVOUS SYSTEM & REFLEXES

The autonomic nervous system controls involuntary functions of viscera, glands, and smooth and cardiac muscle. Unlike the somatic system, it uses a two-neuron chain: a myelinated preganglionic neuron synapses in a ganglion onto an unmyelinated postganglionic neuron.

Feature	Sympathetic	Parasympathetic
Outflow	Thoracolumbar (T1 to L2)	Craniosacral (CN III, VII, IX, X; S2 to S4)
Preganglionic neuron	Short (paravertebral or prevertebral ganglia)	Long (ganglia in or near the target organ)
Postganglionic transmitter	Noradrenaline (alpha and beta receptors)	Acetylcholine (muscarinic M1 to M5)
Role	Fight or flight (catabolic)	Rest and digest (anabolic)
Exceptions	Sweat glands are cholinergic; adrenal medulla uses ACh	-

Organ	Sympathetic	Parasympathetic
Heart	Raises rate and contractility (beta-1)	Lowers rate (M2)
Bronchi	Dilate (beta-2)	Constrict (M3)
GI tract	Inhibits motility; closes sphincters	Stimulates peristalsis; relaxes sphincters
Bladder	Relaxes detrusor; closes sphincter (retention)	Contracts detrusor (urination)
Pupil	Dilates (alpha-1, mydriasis)	Constricts (M3, miosis)

4.1 Reflexes

Reflex	Arc	Clinical note
Stretch (deep tendon)	Monosynaptic; Ia afferent to alpha-motor neuron	Knee (L3/4), ankle (S1); absent in LMN, brisk in UMN lesions
Golgi tendon (inverse)	Disynaptic; Ib inhibitory interneuron	Clasp-knife response in spasticity
Withdrawal (flexor)	Polysynaptic, multisegmental	Protective; crossed extension supports the opposite limb
Babinski	Polysynaptic	Hallux dorsiflexion is abnormal (UMN sign) after age 2
Pupillary light	CN II afferent, CN III efferent	Tests direct and consensual response; RAPD in afferent defect

5 VISION

The special senses are specialised systems in the head, each served by a dedicated cranial nerve and a specialised cortical area: vision (CN II, occipital cortex), hearing and balance (CN VIII, temporal cortex), smell (CN I, which uniquely bypasses the thalamus), and taste (CN VII, IX, X, insula). This section covers vision; the next covers the others.

5.1 Anatomy of the eye

Structure	Function
Cornea	Transparent, avascular; provides ~70% of refractive power (~43 D)
Sclera	White fibrous protective shell; attachment for extraocular muscles
Ciliary body	Contracts to round the lens (accommodation); produces aqueous humour
Iris	Controls pupil size (dilator alpha-1, constrictor M3)
Retina	Phototransduction; ganglion cell axons form the optic nerve
Fovea	Central pit of the macula; cones only; greatest visual acuity
Aqueous humour	Maintains IOP (10 to 21 mmHg); drains via trabecular meshwork to Schlemm canal
Lens	Biconvex, avascular; ~20 D adjustable; opacity is cataract

5.2 Optics and refractive errors

The eye has a total power of about +60 D at infinity (cornea ~43 D fixed, lens ~20 D variable). Light must converge exactly on the fovea.

Error	Problem	Focus	Correction
Emmetropia	None	On the retina	None
Myopia	Eye too long / too curved	In front of retina	Concave (diverging) lens
Hypermetropia	Eye too short / too flat	Behind retina	Convex (converging) lens
Astigmatism	Unequal curvature	Multiple focal points	Cylindrical lens
Presbyopia	Age-related loss of lens elasticity	Cannot focus near	Reading glasses (convex)

The accommodation triad (near response, via CN III)

Focusing on a near object triggers three simultaneous responses: accommodation (ciliary muscle contracts, lens rounds, power increases), pupillary constriction (raises depth of field), and convergence (both eyes turn medially). Raised trabecular resistance instead raises IOP and causes open-angle glaucoma, with progressive retinal ganglion cell and visual field loss.

5.3 Photoreception and phototransduction

Property	Rods	Cones
Number	~120 million	~6 to 7 million
Location	Peripheral retina	Fovea (highest density)
Sensitivity	Very high; dim light (scotopic)	Lower; bright light (photopic)
Colour	None (rhodopsin)	Trichromatic (blue, green, red opsins)
Acuity	Low (high convergence)	High (1:1 at the fovea)

In the dark, cGMP is high, cation channels are open, and the photoreceptor is depolarised, releasing glutamate continuously. A photon isomerises 11-cis retinal in rhodopsin to all-trans, activating transducin, which activates phosphodiesterase; cGMP falls, the channels close, and the cell hyperpolarises and reduces glutamate release. Recovery involves rhodopsin kinase, arrestin, and resynthesis of cGMP.

Colour vision and colour blindness

Colour is coded from the ratio of activation of long, medium, and short cones through opponent channels. Red-green colour blindness (about 8% of males, X-linked) is the commonest form, from loss or shift of the L (protan) or M (deutan) pigment. Complete cone failure is achromatopsia.

5.4 The visual pathway

Retinal ganglion cell axons form the optic nerve. At the optic chiasm the nasal fibres (which carry the temporal visual field) cross, while temporal fibres stay ipsilateral, so each optic tract carries the contralateral visual field from both eyes. Fibres relay in the lateral geniculate nucleus and travel via the optic radiation to the primary visual cortex (V1) around the calcarine sulcus.

Lesion site	Visual field defect	Typical cause
Optic nerve	Unilateral blindness (that eye)	Optic neuritis (MS)
Optic chiasm (central)	Bitemporal hemianopia	Pituitary adenoma
Meyer loop (temporal)	Contralateral superior quadrantanopia	Temporal lobe lesion
Occipital cortex	Contralateral homonymous hemianopia with macular sparing	PCA infarction

5.5 Eye movements

Saccades are rapid voluntary shifts of gaze (frontal eye fields); smooth pursuit tracks a moving target; vergence aligns the eyes for near objects; the vestibulo-ocular reflex stabilises gaze during head movement. All use CN III, IV, and VI via brainstem gaze centres.

Internuclear ophthalmoplegia (INO)

The medial longitudinal fasciculus links the CN VI nucleus to the contralateral CN III nucleus to coordinate horizontal gaze. An MLF lesion causes failure of adduction in the ipsilateral eye on lateral gaze, with nystagmus of the abducting eye. Bilateral INO in a young person is multiple sclerosis until proven otherwise.

6**HEARING, BALANCE, SMELL & TASTE****6.1 The ear**

Part	Structures	Function
External ear	Pinna, ear canal, tympanic membrane	Collects and channels sound; TM vibrates
Middle ear	Ossicles (malleus, incus, stapes), Eustachian tube	Impedance matching (~22x) between air and cochlear fluid
Inner ear (cochlea)	Three fluid chambers; organ of Corti on the basilar membrane	Converts vibration to electrical signals; tonotopic frequency analysis
Vestibular apparatus	Semicircular canals, utricle, saccule	Detects angular and linear acceleration and gravity

The basilar membrane is tonotopic: its stiff narrow base responds to high frequencies and its wide flexible apex to low frequencies (the travelling wave). Outer hair cells actively amplify the wave via prestin, sharpening frequency selectivity.

6.2 Hair cell transduction and the auditory pathway

Basilar membrane movement shears the stereocilia against the tectorial membrane; deflection toward the tallest row stretches tip links and opens transduction channels. Potassium-rich endolymph (the +80 mV endocochlear potential) flows in, depolarising the hair cell; calcium then triggers glutamate release onto spiral ganglion neurons, generating action potentials in CN VIII. The pathway ascends through the cochlear nuclei, superior olivary complex (sound localisation), inferior colliculus, and medial geniculate nucleus to the primary auditory cortex. Bilateral representation means unilateral cortical lesions rarely cause deafness.

6.3 Disorders of hearing

Type	Rinne	Weber	Causes
Conductive	Bone > air (abnormal)	Lateralises to the affected ear	Wax, perforation, otitis media, otosclerosis
Sensorineural	Air > bone (both reduced)	Lateralises to the unaffected ear	Noise, presbycusis, ototoxic drugs, acoustic neuroma

6.4 Vestibular system and equilibrium

The three semicircular canals detect angular (rotational) acceleration: endolymph lags behind head rotation and deflects the cupula, bending hair cells in the ampulla. The utricle (horizontal) and saccule (vertical) detect linear acceleration and gravity, using otoconia that slide over the macula. Signals travel via CN VIII to the vestibular nuclei, which connect to the cerebellum, spinal cord (posture), eye muscles (VOR), cortex, and autonomic centres (motion sickness).

Nystagmus: peripheral versus central

Peripheral (labyrinth or CN VIII): horizontal or torsional, fatigable, unidirectional, suppressed by visual fixation, with prominent vertigo. Causes include BPPV, vestibular neuritis, and Meniere disease. Central (brainstem or cerebellum): pure vertical or direction-changing, non-fatigable, not suppressed by fixation, with neurological signs. Causes include stroke, MS, and tumour.

6.5 Olfaction (smell)

About 10 to 20 million bipolar olfactory receptor neurons in the superior nasal cavity each express one of roughly 400 receptor genes (GPCRs signalling through cAMP). Axons form CN I and synapse in glomeruli of the olfactory bulb, where neurons expressing the same receptor converge. Uniquely, olfaction projects directly to the cortex (piriform and entorhinal) and the amygdala and hippocampus without a thalamic relay, which explains the strong emotional and memory associations of smell. Olfactory neurons are replaced every 30 to 60 days.

Anosmia: clinical significance

Loss of smell may indicate sinonasal disease (commonest, reversible), head trauma (shearing at the cribriform plate), early Parkinson or Alzheimer disease (among the earliest changes), Kallmann syndrome (congenital anosmia with hypogonadotropic hypogonadism), or viral infection such as COVID-19.

6.6 Gustation (taste)

Around 10,000 taste buds sit on tongue papillae, palate, and epiglottis, each holding 50 to 150 taste receptor cells. There are five primary tastes. Sweet (T1R2+T1R3) and umami (T1R1+T1R3) and bitter (T2R family) act through GPCRs and gustducin; sour is sensed by proton channels (OTOP1) on type III cells; salty is sensed by ENaC sodium channels. Taste travels via CN VII (anterior two-thirds of tongue), CN IX (posterior third), and CN X (palate and epiglottis) to the nucleus tractus solitarius, then the thalamus (VPMpc, uncrossed) and the gustatory cortex in the insula.

7**RENAL PHYSIOLOGY: THE KIDNEY & NEPHRON**

The kidneys are paired retroperitoneal organs (T11 to L3, 11 to 15 cm long) that maintain homeostasis by filtering blood and forming urine. The cut surface shows an outer cortex (glomeruli, PCT, DCT) and an inner medulla of 8 to 18 pyramids whose papillae drain into minor calyces, then major calyces, the renal pelvis, and the ureter.

7.1 Functions of the kidney

- Excretory: removes metabolic wastes, drugs, and toxins (basis of renal function tests).
- Homeostatic: water balance, electrolyte balance (Na, K, Ca, phosphate), and acid-base balance (secretes H⁺, reabsorbs bicarbonate).
- Endocrine: renin (RAAS), erythropoietin (red cell production), and prostaglandins.
- Metabolic: activates vitamin D to calcitriol via 1-alpha-hydroxylase in the PCT, under PTH control, promoting intestinal calcium absorption.

7.2 The nephron

The nephron is the functional unit, about 1.2 million per kidney, made of a renal corpuscle (filter) and a long tubule. Filtrate flows: Bowman space, proximal convoluted tubule, loop of Henle, distal convoluted tubule, collecting duct, papillary duct, calyces, pelvis, ureter.

Renal corpuscle and the filtration barrier

Barrier layer	Property
Fenestrated endothelium	Pores 70 to 90 nm; blocks blood cells
Basement membrane	Negatively charged gel; excludes particles above 8 nm and repels albumin
Filtration slits (podocytes)	Allow particles below 3 nm; negatively charged

Water, electrolytes, glucose, amino acids, and nitrogenous wastes pass freely; plasma proteins and protein-bound substances are retained. Damage to the barrier causes proteinuria and haematuria.

Renal tubule segments

Segment	Key features
Proximal convoluted tubule	Brush border, many mitochondria; reabsorbs ~65% of filtrate
Loop of Henle	Descending limb permeable to water; thick ascending limb pumps NaCl but is water-impermeable
Distal convoluted tubule	Cuboidal, no microvilli; end of the nephron proper
Collecting duct	Principal cells (aldosterone, ADH, K secretion); intercalated cells (acid-base)

Cortical nephrons (85 to 90%) have short loops; juxtamedullary nephrons (10 to 15%) have long loops served by the vasa recta and are essential for concentrating urine.

7.3 Renal circulation

The kidneys are 0.4% of body weight but receive about 25% of cardiac output (1300 mL/min), not for metabolism but to support filtration. Blood passes through two capillary beds in series: high-pressure glomerular capillaries (filtration) and low-pressure peritubular capillaries or vasa recta (reabsorption), separated by the efferent arteriole. Sympathetic fibres regulate arteriolar tone and stimulate renin when blood pressure falls.

7.4 Glomerular filtration and GFR

Net filtration pressure

NFP = blood hydrostatic pressure (60 mmHg out) minus capsular pressure (18 mmHg in) minus colloid osmotic pressure (~32 mmHg in) = about 10 mmHg outward. GFR = NFP times the filtration coefficient K (~12.5 mL/min per mmHg). Normal GFR is about 125 mL/min (180 L/day) in men and about 105 mL/min in women.

GFR must be tightly controlled: too high causes dehydration and electrolyte loss; too low causes azotemia. It is regulated by three mechanisms. Renal autoregulation keeps GFR stable over a MAP of 80 to 170 mmHg through the myogenic response (stretch causes afferent constriction) and tubuloglomerular feedback (macula densa senses NaCl and adjusts afferent tone). Sympathetic control constricts the afferent arteriole in shock or exercise. Hormonal control through the renin-angiotensin system raises blood pressure: renin converts angiotensinogen to angiotensin I, ACE converts it to angiotensin II, which causes vasoconstriction, aldosterone release, and thirst.

8

URINE FORMATION, CONCENTRATION & CLINICAL CORRELATES

Of the 180 L/day of filtrate, about 99% is reabsorbed, leaving 1 to 2 L of urine. Three processes transform plasma into urine: filtration (corpuscle), reabsorption (tubule to blood), and secretion (blood to tubule).

8.1 Tubular reabsorption

Reabsorption uses transcellular (through cells) and paracellular (between cells) routes. In the PCT, sodium is the key driver: its reabsorption creates gradients that pull water and solutes, and 65% of filtrate is reclaimed here (glucose and amino acids fully via SGLT co-transport). The loop of Henle builds the medullary osmotic gradient.

The DCT and collecting duct are hormonally regulated.

Hormone	Action
Aldosterone	Raises Na reabsorption and water retention; lowers urine volume (triggered by angiotensin II)
ADH (vasopressin)	Inserts aquaporin-2 in the collecting duct; water reabsorbed; concentrated urine
ANP	Released when BP rises; inhibits RAAS and NaCl reabsorption; raises salt and water excretion
PTH	Raises Ca reabsorption in the DCT; raises phosphate excretion in the PCT; stimulates calcitriol

About 85% of water reabsorption is obligatory (follows solute in the PCT and loop, regardless of hydration); the remaining 15 to 18% is facultative, occurring in the collecting duct under ADH control.

8.2 Tubular secretion

Secretion moves wastes and excess ions from peritubular blood into the tubule. It removes urea, uric acid, ammonia, creatinine, and many drugs (penicillin, aspirin, morphine), which is why many drugs need frequent dosing. It also maintains acid-base balance: H^+ secreted in the PCT regenerates bicarbonate, and intercalated cells in the DCT and collecting duct secrete H^+ to acidify urine.

8.3 Handling of specific substances

Substance	Handling
Glucose	Fully reabsorbed in the PCT via SGLT; renal threshold ~180 mg/dL; T_m ~375 mg/min, beyond which glycosuria occurs
Sodium	Master solute; reabsorbed throughout the nephron; drives obligatory water reabsorption
Potassium	Mostly reabsorbed early; secreted in DCT/CD under aldosterone, plasma K, flow, and pH
Urea	Recycled; inner medullary collecting duct reabsorbs urea (UT-A, ADH-stimulated) to boost the gradient
PAH	Filtered and secreted; used to measure renal plasma flow

8.4 Concentration of urine and the countercurrent mechanism

The kidney varies urine concentration using the medullary osmotic gradient (300 mOsm/L in the cortex up to 1200 mOsm/L at the papilla). Countercurrent multiplication generates this gradient: the thick ascending limb pumps NaCl out but is water-impermeable, making the interstitium hypertonic, while the water-permeable descending limb loses water and concentrates its fluid. The vasa recta perform countercurrent exchange, preserving the gradient while nourishing the medulla.

ADH and the control of water loss

Dehydration raises blood osmolarity, so the hypothalamus drives ADH release; aquaporins are inserted in the collecting duct, water is reabsorbed, and concentrated urine (up to 1200 mOsm/L) is produced. Overhydration lowers osmolarity, ADH falls, the collecting duct stays water-impermeable, and dilute urine (as low as 50 to 65 mOsm/L) results. Tubular fluid entering the distal tubule is always hypotonic because the thick ascending limb removes solute without water.

8.5 Composition and properties of urine

Property	Value
Volume	1 to 2 L/day (below 400 mL/day risks azotemia)
pH	4.5 to 8.2 (usually ~6.0, mildly acidic)
Specific gravity	1.001 to 1.028
Osmolarity	50 to 1200 mOsm/L (blood ~300)
Composition	95% water, 5% solutes (urea most abundant, then creatinine, uric acid, ions)

Glucose, albumin, free haemoglobin, ketones, and bile pigments are abnormal in urine. Volume terms: polyuria (above 2 L/day), oliguria (below 500 mL/day), and anuria (0 to 100 mL/day).

8.6 Clinical correlates

Condition	Mechanism
Diabetes mellitus / glycosuria	Plasma glucose exceeds the renal threshold, saturating SGLT; osmotic diuresis and polyuria
Diabetes insipidus	ADH deficiency (central) or resistance (nephrogenic); massive dilute urine, no glucose
Proteinuria and haematuria	Filtration barrier damage lets protein and red cells into urine
Azotemia	Urine output below 400 mL/day; nitrogenous wastes accumulate; a marker of renal failure
Hemodialysis	Artificial filtration in end-stage disease; wastes diffuse across a semipermeable membrane



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How to use this guide

Work through one section at a time. Read the prose for understanding, then use the tables and callout boxes for rapid revision and last-minute recall. Coloured boxes flag core principles, clinical pearls, drug targets, and emergencies. Every fact here has been rewritten in original wording from primary physiology teaching material.